

Node.js Path Module

Professional Classroom Notes - File & Directory Path Utilities

Learning objectives: Understand why the built-in path module is used and correctly apply `join()`, `resolve()`, `basename()`, `extname()`, `dirname()`, `parse()`, `sep`, and the practiced extra method `isAbsolute()`.

1. What is the Path Module?

The Node.js **path** module is a built-in module used to work with file and directory paths. It helps construct paths, extract file information, resolve absolute paths, and handle operating-system path separators.

```
const path = require("path");
```

Important: Because path is built into Node.js, no separate npm installation is required.

2. path.join()

Purpose: Combines path segments into one normalized path.

```
const filePath = path.join("projects", "nodejs", "backend", "server.js");
console.log(filePath);
```

Depending on the operating system, the separator may appear differently. The method should be preferred over manually joining path pieces with hard-coded slashes.

3. path.resolve()

Purpose: Resolves the supplied path segments into an absolute path. A relative path can therefore be converted into a full path based on the current working location.

```
const absolutePath = path.resolve(filePath);
console.log(absolutePath);
```

join vs resolve: `join()` mainly combines segments; `resolve()` produces an absolute resolved path.

4. path.basename()

Purpose: Returns the final portion of a path, commonly the file name.

```
const fileName = path.basename(filePath);
console.log(fileName); // server.js
```

5. path.extname()

Purpose: Returns the extension of the final file portion.

```
const extension = path.extname(filePath);
console.log(extension); // .js
```

6. path.dirname()

Purpose: Returns the directory portion of a path, excluding the final file name.

```
const directory = path.dirname(filePath);
console.log(directory);
```

7. path.parse()

Purpose: Breaks a path into a structured object. This is useful when several pieces of file information are required at once.

```
const parsedPath = path.parse(filePath);
console.log(parsedPath);
```

Important properties returned by `parse()`:

Property	Meaning	Example
root	Root of the path	/ or drive root
dir	Directory path	projects/.../backend
base	File name with extension	server.js
ext	File extension	.js
name	File name without extension	server

```
console.log(parsedPath.name); // server
console.log(parsedPath.base); // server.js
console.log(parsedPath.ext); // .js
console.log(parsedPath.dir); // directory path
```

8. path.sep

Purpose: Gives the platform-specific path separator.

```
console.log(path.sep);
```

This helps explain why a path may use / on one operating system and \ on another.

9. path.isAbsolute() - Extra Practice

Purpose: Checks whether a path is absolute. It returns a Boolean value: true or false.

```
const result = path.isAbsolute(filePath);
console.log(result);
```

10. Complete Classroom Example

```
const path = require("path");

const filePath = path.join("projects", "nodejs", "backend", "server.js");
const absolutePath = path.resolve(filePath);
const fileName = path.basename(filePath);
const extension = path.extname(filePath);
const directory = path.dirname(filePath);
const parsedPath = path.parse(filePath);

console.log("Full Path:", filePath);
console.log("Absolute Path:", absolutePath);
console.log("File Name:", fileName);
console.log("Extension:", extension);
console.log("Directory:", directory);
console.log("Name:", parsedPath.name);
```

```
console.log("Base:", parsedPath.base);
console.log("Parsed Extension:", parsedPath.ext);
console.log("Parsed Directory:", parsedPath.dir);
console.log("Separator:", path.sep);
console.log("Is Absolute:", path.isAbsolute(filePath));
```

11. Quick Comparison

Feature	Main Job
join()	Combine path segments
resolve()	Create/resolve an absolute path
basename()	Get final file/folder name
extname()	Get file extension
dirname()	Get directory portion
parse()	Break path into an information object
sep	Get OS path separator
isAbsolute()	Check whether a path is absolute

12. Solved Practice

Question: For the path `students/junaid/nodejs/practice.js`, find the file name, extension, directory, parsed name and parsed base.

```
const path = require("path");
const fp = path.join("students", "junaid", "nodejs", "practice.js");
const info = path.parse(fp);

console.log(path.basename(fp)); // practice.js
console.log(path.extname(fp)); // .js
console.log(path.dirname(fp));
console.log(info.name);        // practice
console.log(info.base);        // practice.js
```

13. Student Practice Tasks

Task A: Build `school/students/results/final.pdf` with `join()`. Print its `basename`, `extension`, `directory`, `parsed name`, `parsed base`, `separator`, and `resolved absolute path`.

Task B: Build `app/assets/images/logo.png`. Use `parse()` once, store the returned object in a variable, and print its `dir`, `base`, `ext`, and `name` separately.

Concept check: Explain in your own words why array/object data is not involved in `path.parse()` input, but its returned result is an object.

14. Oral / Revision Questions

1. Why do we use the Path module?
2. What is the difference between `join()` and `resolve()`?
3. Which method returns only the extension?
4. Which method returns the directory?
5. What does `basename()` return?

6. What type of value does `parse()` return?
7. What is the difference between `parse().name` and `parse().base`?
8. What does `path.sep` represent?
9. What does `isAbsolute()` return?
10. Why is using path utilities better than manually assuming one path separator?